

Biology, 2017 EC, entrance exam, version 2

Grade 12 · Biology · 2017 E.C.

Time allowed: 120 minutes

Questions: 96

Total marks: 96

Name _____ Grade _____ ID No. _____ Date _____

1. The vaccine against terrible cattle disease rind pest was made by _____. (1)
 - A. Prof. Beyene Petros
 - B. Prof. Tilahun Yilma
 - C. Dr. Alipurduar Lemma
 - D. Dr. Gebissa Ejetta
2. Which of the following Ethiopian Biologist has been honored by developing high yielding strains of sorghum? (1)
 - A. Dr. Tsehaynesh Meselle
 - B. Dr. Birhane Asfaw
 - C. Dr. Zeresenay Alemseged
 - D. Dr. Gebissa Ejetta
3. Ethiopian Biologist who worked on effects of chemical pesticides on human is _____. (1)
 - A. Prof. Sebsibie Demisew
 - B. Prof. Legesse Negash
 - C. Prof. Yealemtsehay Mekonnen
 - D. Prof. Mogessie Ashenafi
4. Ethiopian Biologist who has made a soap berry to eradicate snail which is intermediate host of bilharzia: (1)
 - A. Dr. Melaku Werede
 - B. Dr. Gebissa Ejetta
 - C. Dr. Birhane Asfaw
 - D. Dr. Aklilu Lemma
5. Among the following renowned Ethiopian Biologists who discover two 160,000 years old human skulls which have a big impact on the study of human evolution? (1)
 - A. Dr. Berhane Asfaw
 - B. Dr. Tsehaynesh Meselle
 - C. Dr. Zeresenay Alemseged
 - D. Dr. Melaku Werede
6. All are the major research areas of EHNRI except: (1)
 - A. HIV vaccine trial
 - B. Forestry
 - C. Battle against TB
 - D. Nutritional state of mothers and baby
7. Ethiopian Biological research center which is involved in mechanized farming is: (1)
 - A. EIAR
 - B. ALIPB
 - C. AHRI
 - D. IBC
8. Ethiopian Biological research center which is highly devoted on research related to Leprosy is: (1)
 - A. IAR
 - B. AHRI
 - C. IBC
 - D. EHNRI
9. Among the following institutes which one is working on the management of the ecosystem? (1)
 - A. ALIPB
 - B. EHNRI
 - C. IBC
 - D. EIAR
10. The resolution power of a microscope used to: (1)
 - A. Show objects clearly
 - B. Increase the size of objects
 - C. Make objects brighter
 - D. Make objects look smaller

-
11. Preparation of specimen to see under the microscope is known as _____. (1)
- | | |
|---------------|------------------|
| A. Resolution | B. Mounting |
| C. Staining | D. Magnification |
-
12. Cells that are adapted to carry out a particular body function are: (1)
- | | |
|-------------------------|---------------------------|
| A. Embryonic stem cells | B. Undifferentiated cells |
| C. Specialized cells | D. Unspecialized cells |
-
13. A cell with a large number of mitochondria is expected to: (1)
- | | |
|-----------------------------------|------------------------------------|
| A. Produce more digestive enzymes | B. Perform a lot of photosynthesis |
| C. Store more food | D. Make a lot of ATP |
-
14. Which of the following structures of the cell are involved in the synthesis of protein? (1)
- | | |
|----------------|-------------|
| A. Chloroplast | B. Ribosome |
| C. Golgi body | D. Vacuole |
-
15. Which of the following organelle is the site of photosynthesis? (1)
- | | |
|----------------|---------------|
| A. Lysosome | B. Centriole |
| C. Chloroplast | D. Leucoplast |
-
16. The movement of water molecules from a region of high concentration to low concentration is: (1)
- | | |
|----------------------|---------------------|
| A. Osmosis | B. Diffusion |
| C. Passive transport | D. Active transport |
-
17. Which of the following is true about rate of diffusion? (1)
- | |
|---|
| A. Diffusion increases when the temperature decreases |
| B. Diffusion increases when the temperature increases |
| C. Diffusion decreases when there is very high concentration difference between two areas |
| D. Diffusion increases when there is very low concentration difference between two areas |
-
18. If an animal cell is put in concentrated salt solution which of the following event will happen? (1)
- | | |
|--------------------------------|------------------------------|
| A. The cell became plasmolysis | B. Water moves into the cell |
| C. The cell became shrunk | D. Salt moves into the cell |
-
19. The most common monomer of carbohydrate is a molecule of: (1)
- | | |
|------------------|------------------|
| A. Phospholipids | B. Fatty acids |
| C. Glucose | D. Triglycerides |
-
20. When 50 amino acids joined together through condensation reaction, how many peptide bonds are formed and water molecules are released respectively? (1)
- | | |
|-----------|-----------|
| A. 48, 50 | B. 49, 49 |
| C. 50, 49 | D. 49, 50 |
-
21. Unsaturated fatty acids are differ from saturated fatty acids except: (1)
- | | |
|-----------------------------------|---|
| A. They have all single bonds | B. They have one or more double bonds |
| C. They are more common in plants | D. They are liquids at room temperature |

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- 22.** Which of the following minerals is important constituent of bone and teeth needed for many metabolic processes? (1)
- A. Iron
B. Calcium
C. Sodium
D. Phosphorus
-
- 23.** One of the following vitamins is needed for the formation of connective tissues which holds the body together? (1)
- A. Vitamin B
B. Vitamin A
C. Vitamin C
D. Vitamin D
-
- 24.** Water is crucial in human body for a number of reasons. Which one is not correct about their functions? (1)
- A. It is a universal solvent in any environment
B. It is involved in the transport of substance around the body
C. It is involved in temperature regulation by loss of heat through sweating
D. It is not involved in any chemical reaction which take place in the body
-
- 25.** Which one of the following is not the part of a balanced diet? (1)
- A. Proteins
B. Lipids
C. Cellulose
D. Carbohydrate
-
- 26.** Which one of the following is orders of digestive system that correctly illustrates from the beginning to the end? (1)
- A. Ingestion → digestion → absorption → assimilation → egestion
B. Digestion → absorption → assimilation → egestion → ingestion
C. Assimilation → egestion → absorption → ingestion → digestion
D. Egestion → assimilation → ingestion → absorption → digestion
-
- 27.** In grade 9 laboratory class, the teacher gives the students a sample of food and asked them to investigate the presence of glucose. What kind of test and result will enable the students to confirm the presence of glucose respectively? (1)
- A. Benedict's test and orange colour
B. Biuret test and purple colour
C. Emulsion test and milky white colour
D. Iodine test and blue black colour
-
- 28.** Which of the following factor affecting enzyme activity without denaturing the enzyme molecule? (1)
- A. Low pH
B. High temperature
C. Low temperature
D. High pH
-
- 29.** The shapes of different teeth in humans are ideally suited to their various functions. Which type of teeth is not correctly matched with their functions? (1)
- A. Premolars → chewing food
B. Canines → chewing food
C. Molars → crushing food
D. Incisors → biting food
-
- 30.** In a Benedict's test for sucrose, why does the test sample first boil with hydrochloric acid? (1)
- A. To reduce Benedict's solution
B. Retest mixture with Benedict solution
C. To neutralize the solution
D. To hydrolyze non-reducing sugar
-
- 31.** Which deficient nutrients are correctly paired with their deficiency diseases? (1)
- A. Protein → Scurvy
B. Iron → Marasmus
C. Carbohydrate → Beri-beri
D. Vitamin A → Night blindness

32. When you breathe out during gas exchange, all of the following happens to the breathing movement except? (1)

- A. The volume of the thorax decrease
B. Pressure inside chest cavity decreases
C. Diaphragm moves upward
D. Pressure inside chest cavity increases

33. The muscle that separate the chest cavity from the abdominal cavity is known as: (1)

- A.** Nasal cavity **B.** Trachoma
- C.** Diaphragm **D.** Epiglottis

34. Which of the following is the correct pathway of breathing from the outer to the inner parts? (1)

- A. Nasal cavity → trachea → bronchus → bronchioles → alveolus
B. Alveolus → bronchus → bronchioles → trachea → nasal cavity
C. Alveolus → bronchioles → bronchus → trachea → nasal cavity
D. Nasal cavity → alveolus → bronchus → trachea → bronchioles

35. While oxygen is carried by the red blood cells, most digested food and wastes are carried by the: (1)

- A. Plasma** **B. White blood cells**
C. Platelets **D. Red blood cells**

36. The absolute maximum amount of air you can take into or breathe out of your lungs is called: (1)

- A.** Tidal volume **B.** Vital capacity
C. Residual volume **D.** Breathing hygiene

37. Which of the following products are formed during anaerobic respiration of one glucose molecule in animal cells? (1)

- A. Ethanol, NAD, 2ATP
B. Lactic acid, CO₂, 2ATP
C. Alcohol, ADP, 2ATP
D. Lactic acid, O₂, CO₂

38. Which of the following is the type of blood vessel that carries deoxygenated blood from the heart to lungs? (1)

- A.** Systematic circulation **B.** Pulmonary vein
- C.** Pulmonary artery **D.** Pulmonary circulation

39. The right side chamber valve which directs the flow of blood by opening and closing is called: (1)

- A. Bicuspid valve
B. Tricuspid valve
C. Mitral valve
D. Vena cava

40. Which one of the following statement is true about pulmonary circulation? (1)

- A. Blood flows from the heart to all around the body and back again
- B. Carry only oxygenated blood from the body
- C. Blood flows from the heart to lungs and back again
- D. It transport oxygenated blood from the heart and back again

41. Which organic molecule in our body has functions as antigens and antibodies of immune system? (1)

- A.** Lipids
- B.** Nucleic acids
- C.** Proteins
- D.** Carbohydrates

42. The blood group of human being which is commonly referred to as 'universal acceptor' is: (1)

- A.** Blood group AB **B.** Blood group O
C. Blood group A **D.** Blood group B

43. All of the following factors that can increase the risks of developing hypertension except? (1)

- A. Reducing salt intake
B. Excessive consumption of alcohol
C. Increasing age
D. Sedentary lifestyle

44. The modern technological instruments are learnt from nature by the process of: (1)

- A. Sketching** **B. Imitating**
C. Photographing **D. Drawing**

45. The relevance and promise of biology as a field of science can be best explained by its application: (1)

- A.** Economic growth and urbanization
- B.** Reducing security problems
- C.** Oil and mineral production
- D.** Agriculture and genetic engineering

46. Among the following which one is incorrect statement about the importance of nature for imitating technological products? (1)

- A. It is the sources of materials that enable to imitating
- B. Used to develop modern technologies that provide services
- C. Used to develop knowledge of problem solving ability
- D. To provide limited resources and techniques for society

47. Which technological products have been made through imitation of termite mound architecture? (1)

- A.** Weaving technology
- B.** New drone design
- C.** Building technology
- D.** Techniques of preserving vaccine

48. Which one of the following are the impact of biology on society and natural world except? (1)

- A. Ensuring food security by increasing productivity
- B. Lack of involvement in application of microbial action
- C. Production of antibiotics and treatment for bacterial infection
- D. Achieving better supply of energy and clean water

49. Among which explanation is correctly concerned about ethical issue in biology? (1)

- A. It refers about rightness or wrongness use of biological discoveries
- B. Concerned about only ethical treatment of plant species
- C. Encourage issues about wrongly uses of technology against human
- D. Focus on encouraging unethical treatment of research animals

50. Which common characteristics of animals that have balanced arrangement of body parts around a central point or axis? (1)

- A.** Multicellularity **B.** Motility
C. Sensitivity **D.** Body symmetry

51. Which one of the following is the common characteristic of invertebrate animals? (1)

- A. They have closed circulatory system
- B. They have an exoskeleton
- C. They are warm-blooded animals
- D. They possess a backbone and internal skeleton

52. The form of animal fertilization which young develop within the female by receiving nourishment from mother's blood through placenta is: (1)

A. Viviparity
B. Oviparity
C. Oviparity
D. Viparity

53. During reproduction, some individual insects develop from unfertilized eggs by the process of: (1)

A. Organogenesis
B. Parthenogenesis
C. Metamorphosis
D. Gastrulating

54. Among which of the following organisms reproduce by external fertilization? (1)

A. Birds
B. Rats
C. Crocodiles
D. Frogs

55. Why temperature is the environmental factor that determine sex during incubation of crocodiles? (1)

A. They have chromosomes
B. They can reproduce by asexually
C. They do not have chromosomes
D. They have amniotic eggs

56. Which mating system of birds those involve with two or more members of the opposite sex during a reproductive season? (1)

A. Polygyny
B. Polygamy
C. Monogamy
D. Polyandry

57. Which developmental processes of germ layers later develop into organs and organ system? (1)

A. Organogenesis
B. Cleavage
C. Blastula
D. Partinogenesis

58. All of the following are considered as a beneficial aspects of insects except? (1)

A. Effective pollinators
B. Sources of useful product and protein
C. Biological control agents
D. Non-edible type of animals

59. Which one of the following is true about harmful aspect of insect in agriculture? (1)

A. Damaging crops and reducing production
B. Used as pollinators in crop reproduction
C. Production of various medicines
D. Production of cosmetics and dyes

60. Which one of the following examples of behavior describes when an animal stops responding to stimulus or cue after a period of repeated exposure? (1)

A. Conditioning
B. Reflex
C. Imprinting
D. Habituation

61. The Ethiopian wolf lives and hunt in packs. During patrolling their habitat, both male and female wolves of the group maintain their area with urine containing pheromone. This pattern of behavior is an example of: (1)

A. Operant behavior
B. Territorial behavior
C. Learned behavior
D. Courtship behavior

62. A quick withdrawal of your hand from a hot plate is an example of: (1)

A. Learned behavior
B. Biological clock
C. Key stimulant
D. Innate behavior

-
- 63.** An increase in blood sugar level causes the pancreas to release the hormone insulin; insulin lowers blood sugar level, restoring the body to its original blood glucose level by converting glucose to glycogen. This is an example of: (1)
- A. Positive feedback mechanism B. Negative feedback mechanism
C. Neutral feedback mechanism D. Homeostatic imbalance
-
- 64.** Which one of the following statement explains the specificity of enzymes? (1)
- A. It can be destroyed by excessive heat
B. It can be used over and over again
C. It catalyze one reaction only
D. Speed up a reaction without being used up
-
- 65.** Which protein structures have three-dimensional arrangement and unique shape of all amino acids in a single polypeptide chain? (1)
- A. Quaternary structure B. Primary structure
C. Tertiary structure D. Secondary structure
-
- 66.** Which enzyme-substrate model that shows a change in shape of active site of an enzyme? (1)
- A. Lock and key model B. Induced-fit model
C. Allosteric enzyme D. Regulatory enzyme
-
- 67.** The type of inhibitor that blocks the binding of the substrate to the active site by competition is: (1)
- A. Non-competitive inhibitor B. Competitive inhibitor
C. Uncompetitive inhibitor D. Irreversible inhibitor
-
- 68.** If an enzyme depends on a cofactor for its activity, then the enzyme become active and functional when it is found in the form of: (1)
- A. Holoenzyme B. Coenzyme
C. Apoenzyme D. Mineral ion
-
- 69.** Which of the following is true about feedback inhibition? (1)
- A. The end product binds to the active site of enzyme
B. The end product binds to the allosteric site of enzyme
C. The shape of the active site is not changed
D. It cannot prevent enzyme performance activity
-
- 70.** Which commercial fermentation uses of enzyme that is important for clarification of fruit juices? (1)
- A. Lipases B. Amylases
C. Protease D. Pectinases
-
- 71.** Which type organic molecules that are mainly derived from vitamins? (1)
- A. Apoenzyme B. Cofactor
C. Coenzyme D. Holoenzyme
-
- 72.** In an enzyme catalyzed reaction, the substance that is used again and again is: (1)
- A. An enzyme B. The substrate
C. The product D. The enzyme-product complex

-
- 73.** The process of heating treatment of germinated grain to dry the green malt and prevent from further germination is known as: (1)
- A. Steeping
B. Kilning
C. Germinating
D. Malting
-
- 74.** Which one of the following involves cutting back the shoots of plants to about 10-15 cm above the ground and allowing the stumps to regrow and produce more branches? (1)
- A. Sawing
B. Grafting
C. Coppicing
D. Cloning
-
- 75.** The state in which all people have physical, social and economical access to sufficient, safe and nutritious food is: (1)
- A. Food security
B. Nutrition security
C. Nutrition science
D. Balanced diet
-
- 76.** All are correct about genetic fingerprinting except: (1)
- A. It is based on RNA analysis
B. It is unique to individuals except identical twins
C. It is made by gel electrophoresis in the PCR machine
D. It is used for crime investigation
-
- 77.** Which one of the following is the characteristic of bacteria? (1)
- A. Their DNA is contained within the cytoplasm of the cell
B. They are larger than eukaryotic cell
C. They have membrane enclosed organelle
D. They have cell wall made up of lipoprotein
-
- 78.** Which one of the following archaea can live in extremely hot temperature? (1)
- A. Acidophiles
B. Hyperthermophiles
C. Psychrophiles
D. Psychrotrophs
-
- 79.** Which one of the following fungi has been used as a test organism to determine Mg, Cu, Zn and Mo in the soil? (1)
- A. *Aspergillus flavus*
B. *Aspergillus parasitic*
C. *Claviceps purpurea*
D. *Aspergillus niger*
-
- 80.** Which one of the following protozoa causes Chagas disease? (1)
- A. *Trypanosoma cruzi*
B. *Trypanosoma brucei*
C. *Toxoplasma gondii*
D. *Plasmodium falciparum*
-
- 81.** All are correct about lysogenic cycle except: (1)
- A. DNA of the virus integrates into the host DNA
B. DNA of the virus doesn't integrate into the host DNA
C. May or may not kill the host cell
D. It takes time
-
- 82.** Which one is the viral symmetry refers to the shape with 20 triangular faces and 12 corners? (1)
- A. Complex
B. Helical
C. Icosahedral
D. Cylindrical

83. Which of the following soil bacteria produce powerful protein toxin that used to kill insect pests? (1)

- A.** *Metarhizium anisopliae*
- B.** *Verticillium lecanii*
- C.** *Beauveria bassiana*
- D.** *Bacillus thuringiensis*

84. All are correct about catabolism except: (1)

- A.** It occurs during cellular respiration
B. It is energy releasing reaction
C. It is destructive phase of metabolism
D. It is energy absorbing reaction

85. Where the light-independent stage of photosynthesis does take place? (1)

- A.** Grana **B.** Thylakoids
C. Lamina **D.** Stroma

86. Which one of the following photosynthetic pigments found only in the photosynthetic members of Chromista and Dinoflagellates? (1)

- A. Chlorophyll 'a'** **B. Chlorophyll 'c'**
C. Carotenoids **D. Phycobilins**

87. All are correct about photosystem II except: (1)

- A. Consists of a pair of Chlorophyll 'a' molecules with an absorption peak at 700nm
- B. Consists of a pair of Chlorophyll 'a' molecules with an absorption peak at 680nm
- C. It passes its electrons to photosystem I
- D. It replenishes its lost electrons from photolysis of water

88. Which one of the following is incorrect about light independent reaction? (1)

- A. It is purely enzymatic
- B. It is slower than the light reaction
- C. It is the first step in the mechanism of photosynthesis
- D. It is the second step in the mechanism of photosynthesis

89. Which one of the following processes occurs during light independent reaction of photosynthesis? (1)

- A.** Change solar energy to chemical energy **B.** Carbon Fixation
C. Splitting of water molecules **D.** Reduced NADP to NADPH

90. The first stage of cellular respiration that takes place in the absence of oxygen is called _____. (1)

- A.** Krebs cycle
B. Electron transport chain
C. Link reaction
D. Glycolysis

91. The amount of net ATP that glycolysis provides to a cell is: (1)

- A. 38
B. 2
C. 36
D. 4

92. In the mitochondrial matrix pyruvic acid reacts with a molecule of coenzyme A to form: (1)

- A. Ethyl alcohol
B. Lactic acid
C. Acetyl CoA
D. NAD

93. A biochemical pathway that breaks down Acetyl CoA is _____. (1)

- A. Krebs cycle
B. Pyruvate oxidation
C. Link reaction
D. Glycolysis

94. The final electron acceptor in the electron transport chain is _____. (1)

- | | |
|-----------------|------------------|
| A. Water | B. Oxygen |
| C. ATP | D. ADP |
-

95. The electron transport chain and the enzyme ATP synthase are embedded in the: (1)

- | | |
|------------------------------|-------------------|
| A. Thylakoid membrane | B. Grana |
| C. Stroma | D. Cristae |
-

96. All of the following occurs during Lactic acid fermentation except? (1)

- A.** Releasing of ethyl alcohol
- B.** Lactic acid accumulates in muscle cells
- C.** Liver converts Lactic acid to pyruvic acid
- D.** NADH oxidized

— END OF EXAMINATION —

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